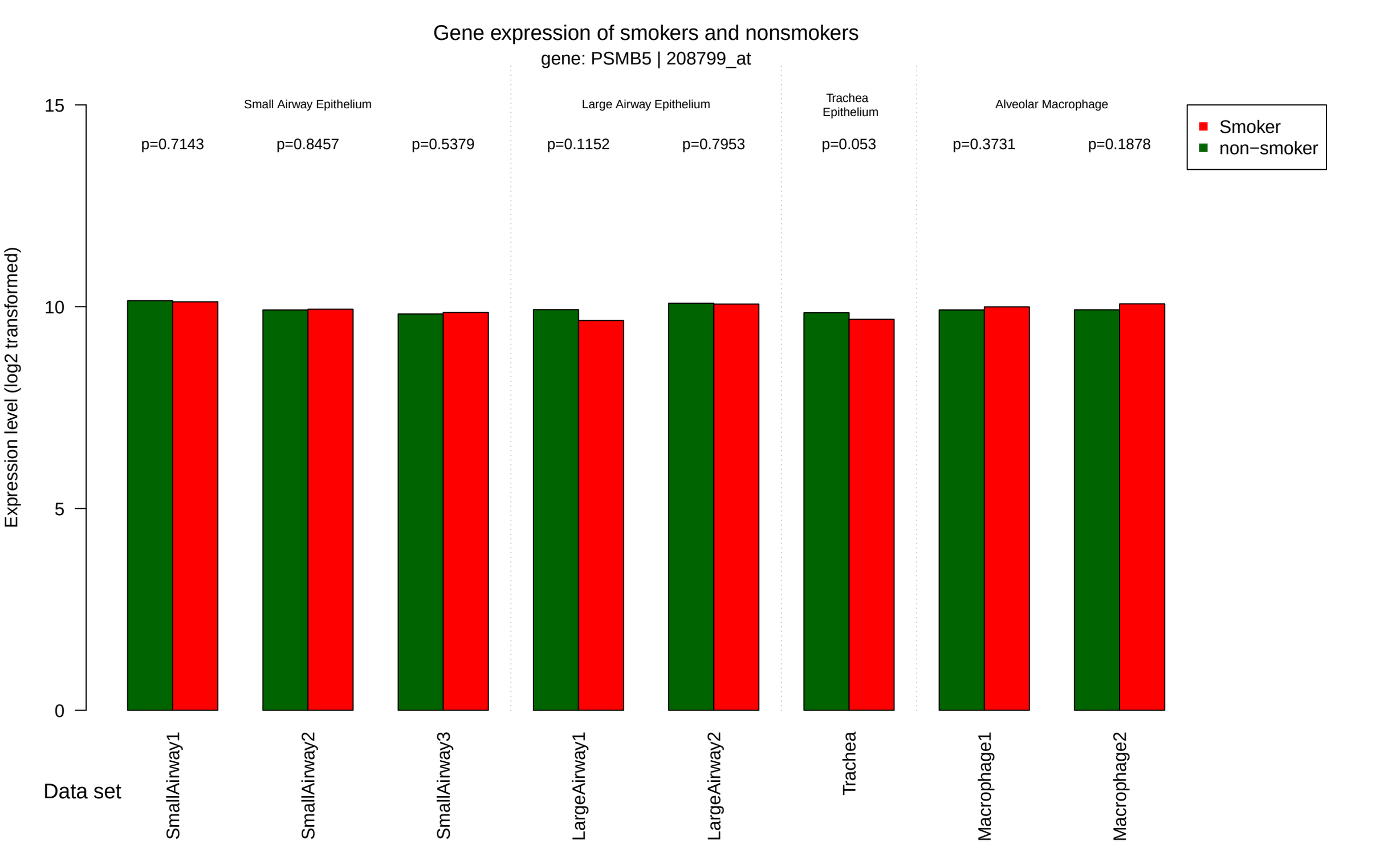
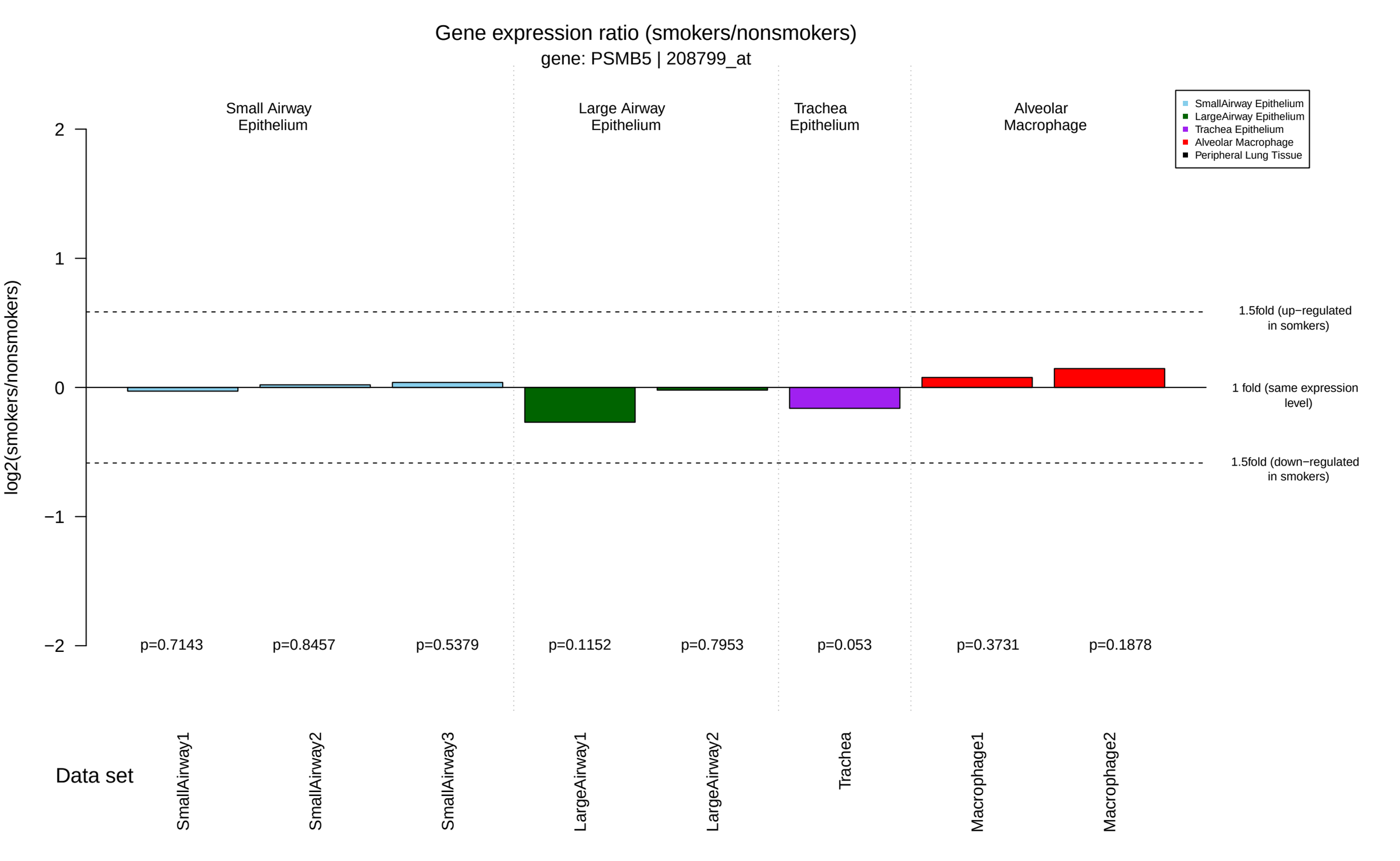
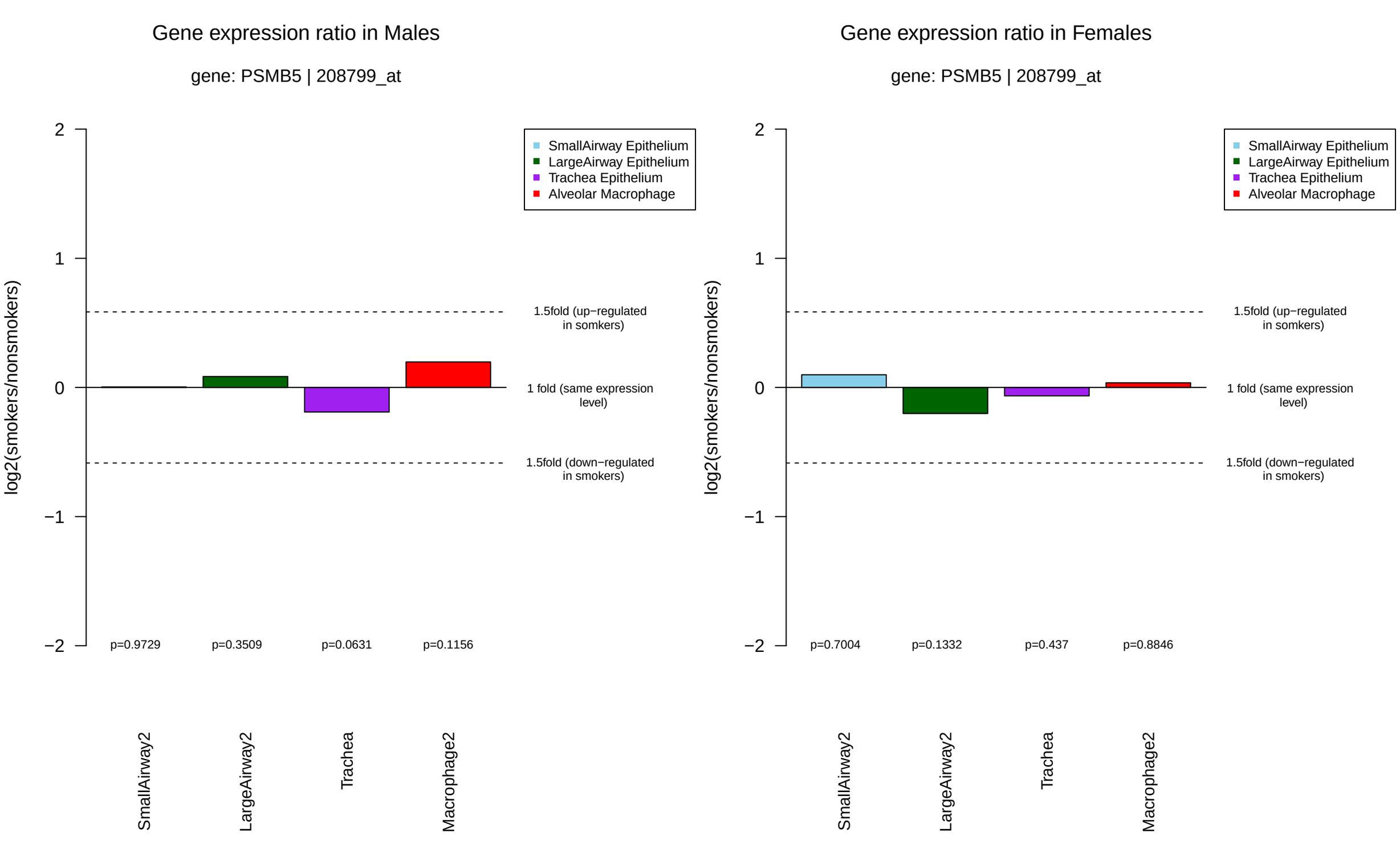
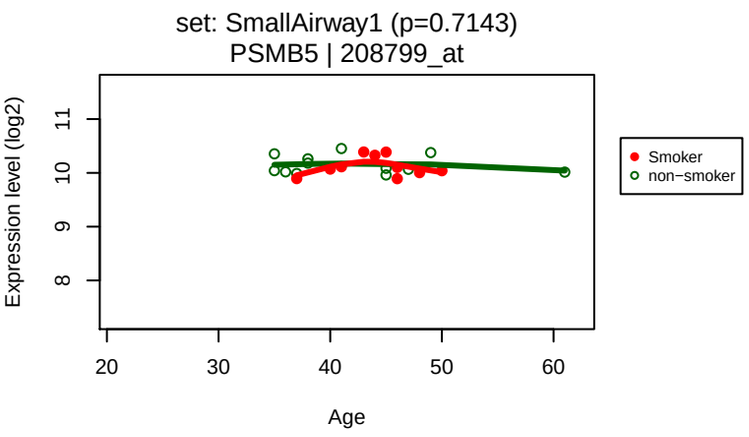
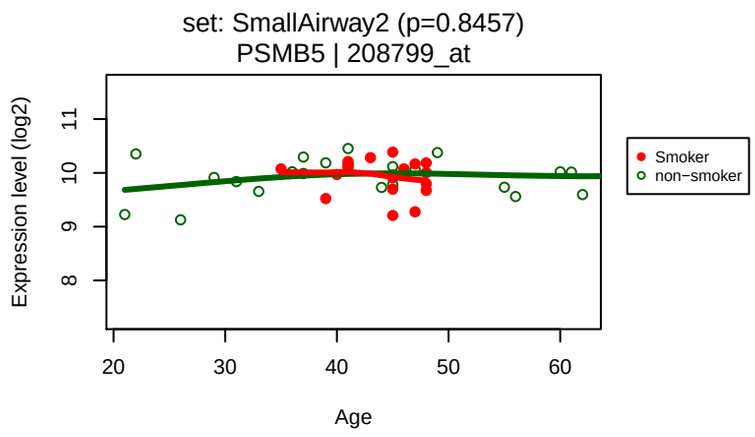
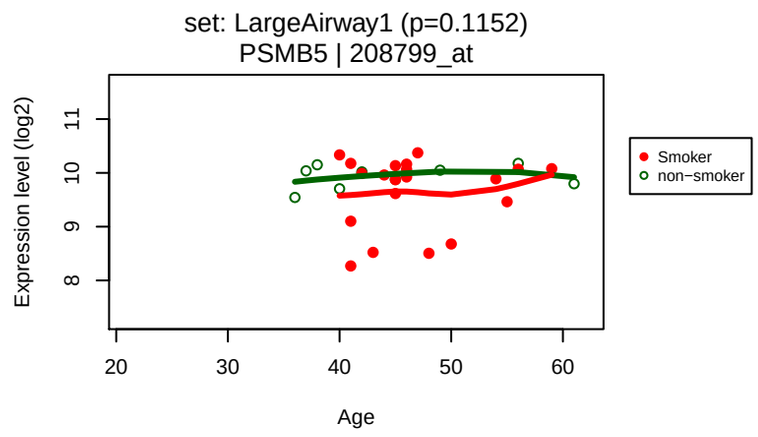

Gene expression of smokers and nonsmokers
gene: PSMB5 | 208799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7143
p=0.8457
p=0.5379
p=0.1152
p=0.7953
p=0.053
p=0.3731
p=0.1878
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

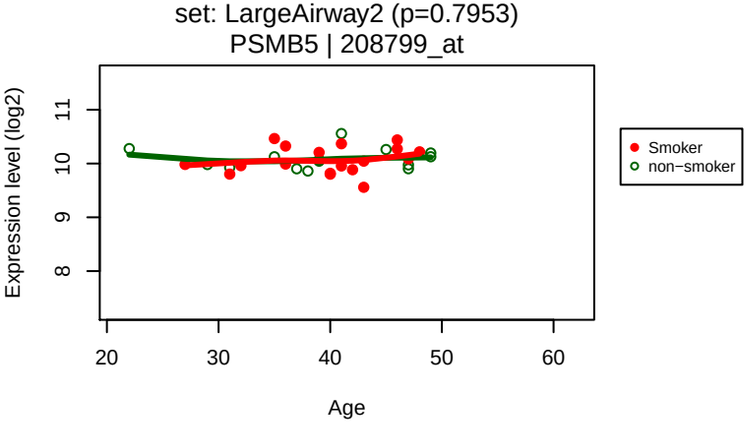

Gene expression ratio (smokers/nonsmokers)
gene: PSMB5 | 208799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7143
p=0.8457
p=0.5379
p=0.1152
p=0.7953
p=0.053
p=0.3731
p=0.1878
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

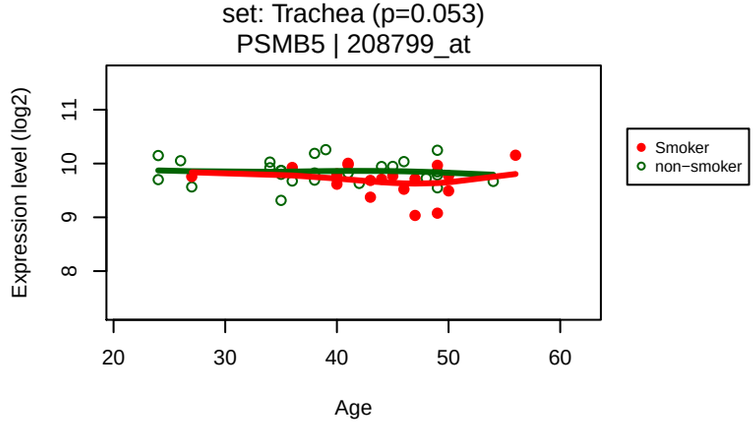

Gene expression ratio in Males
gene: PSMB5 | 208799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9729
p=0.3509
p=0.0631
p=0.1156
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB5 | 208799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7004
p=0.1332
p=0.437
p=0.8846
SmallAirway2
LargeAirway2
Trachea
Macrophage2

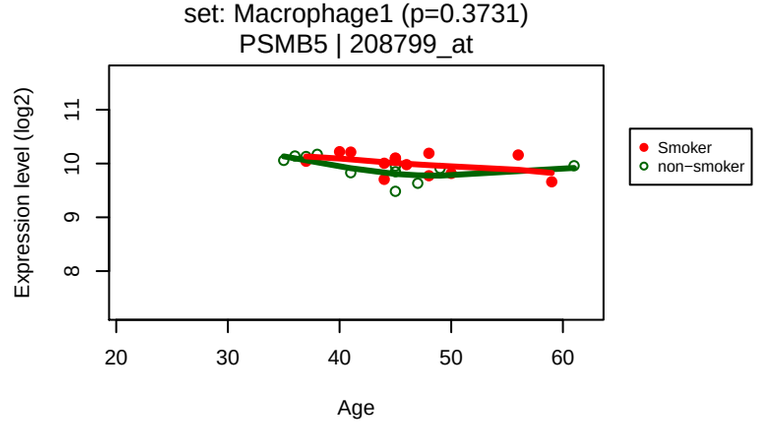

set: SmallAirway1 (p=0.7143)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker

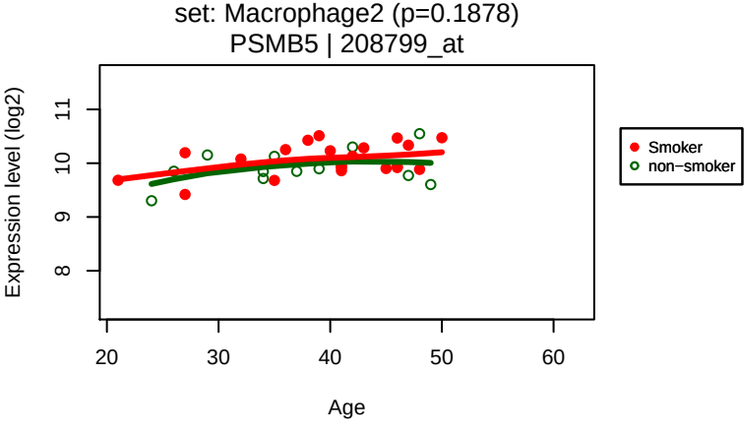

set: SmallAirway2 (p=0.8457)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker

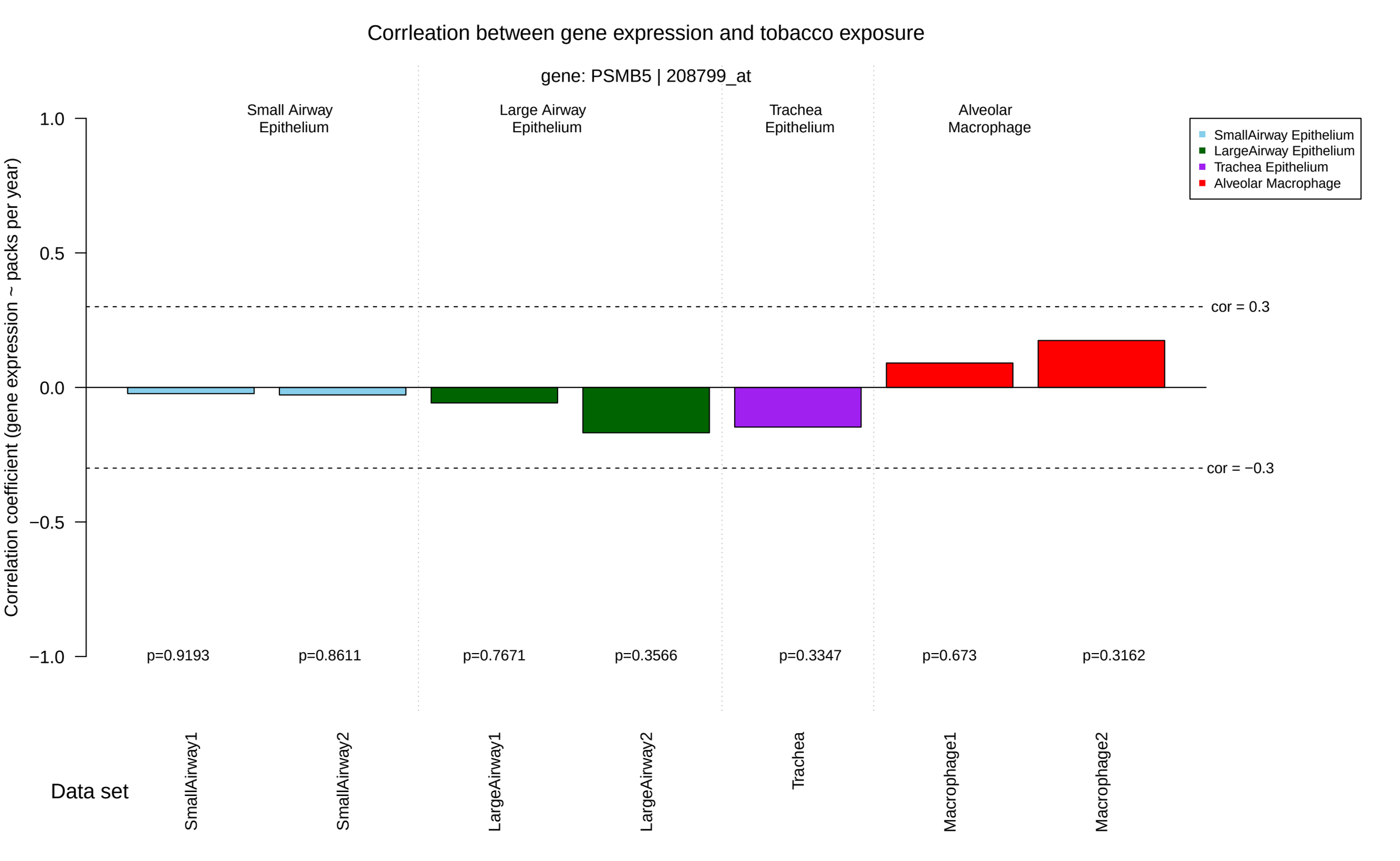

set: LargeAirway1 (p=0.1152)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7953)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.053)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3731)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1878)
PSMB5 | 208799_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB5 | 208799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9193
p=0.8611
p=0.7671
p=0.3566
p=0.3347
p=0.673
p=0.3162
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2